

IT531 Advanced Computer Networks

Instructor: Dr. Abhishek Gupta

Office: 3208, Faculty Block 3

Email: abhishek_gupta@dau.ac.in

Course Placement: Advanced Computer Networks is an elective course offered to fourth year students of the B.Tech program and to M.Tech students.

Course format: It is 3 hours' lecture every week.

Prerequisites: This course assumes knowledge of the first level B.Tech course in Computer Networks. Python and Matlab programming skills (optional) may be useful for the course project.

Course content: Major topics in this course include: Internet protocols for congestion control, approaches to achieve reliability, scalability. Network measurement, P2P networks, Wireless networks, Content delivery networks. Software Defined Networking (SDN) architecture, Network Functions Virtualization (NFV).

This course will emphasize the concepts and issues underlying the design and implementation of the Internet and other kinds of modern wireless networks. We will also spend time learning to quantitatively analyse the performance of network protocols. A combination of reference books and a set of research papers will be used to cover the fundamentals, seminal findings and new directions in networking research. This course will introduce students to the essential tools and concepts such as:

- Understanding a computer networks' basic structure requires concepts.
- Visualize the connected structure where networks are not only limited to computers but span to billions of devices.
- The course aims to develop a working foundation that would allow for systematic analysis of modern networks.
- In addition, it also aims at exposing students to some modern challenges and selected methods that are used extensively in the research literature.
- Understanding the technology of the Internet and learning how to improve its properties is a fundamental objective of this course that will have an enormously beneficial impact on all facets of the economy and society at large.

Course Goals: The goals for this course are (1) To study main concepts of computer network architecture, protocols and systems; (2) To become acquainted with computer networking research topics; (3) To gain practice in reading research papers and discussing the research of others, and (4) To be introduced and gain experience with network programming, analysis of traffic measurements, traffic control algorithms, and peer-to-peer software.

Course Outcomes: At the end of the course, the students should be able to analyse a communication system by separating out the different functions provided by the network; understand that there are fundamental limits to any network; understand the general

principles behind multiplexing, addressing, routing, reliable transmission and other protocols as well as specific examples of each; be able to compare various network architectures and how they solve similar problems with different efficiency; have an informed view of both the internal workings of the Internet and of a number of common Internet applications and protocols. After completing the course, the students would be able to apply the knowledge of computer networks to model complex problems in network engineering (P1,P2,P3,P12), carry out model based investigations to derive insight into the problem at hand (P4), communicate effectively through vivas, course presentation and report writing (P10).

Evaluation and Assessment: There will be assignments, critical review of a research paper and a final exam. The final grade will be based on the assignments (40%), critical review of a research paper (10%) and the final end-semester exam (50%).

The critical review of a research paper aims to gain an understanding of how networking may evolve in the future to provide ubiquitous support for quality-of-service (QoS) in heterogeneous environments. The students will identify and critically review one paper selected from either of the following journals/conferences (not limited to):

- **IEEE/ACM Transactions on Networking**
- **IEEE Transactions on Network Science and Engineering**
- **IEEE Transactions on Network and Service Management**
- **IEEE Transactions on Green Communications and Networking**
- **IEEE Transactions on Wireless Communications**
- **IEEE Wireless Communications and Networking Conference (WCNC)**
- **IEEE Conference on Network Softwarization (NetSoft)**
- **IEEE Conference on Local Computer Networks (LCN)**
- **IEEE Conference on Computer Communications (InfoCom)**

This will help students to get acquainted with some advanced research topics in computer networking, that will enable them to conduct state-of-the-art research in various domains of networking. The final end-semester exam will be a closed book exam and will comprise some multiple choice questions, some mathematical questions and some descriptive questions.

A single textbook is insufficient for this course. Some of the relevant resources that will be referred to are:

Text books:

- Computer Networks. Tanenbaum, A. S., & Wetherall, D. J. 6th ed., Pearson, 2022.
- Computer Networking: A Top-Down Approach. Kurose, James F., and Keith W. Ross. 8th ed., Pearson, 2021.

Reference books:

- Computer Networks: A Systems Approach. Peterson, L. L., & Davie, B. S. 6th ed., Cambridge, MA: Morgan Kaufmann, 2022.

- Stallings, W. (2017). Data and Computer Communications (10th ed.). Pearson Education India.
- Data Communication and Networking, D P Nagpal, S.Chand Publishing, 2011.
- Hassan, Mahboob and Jain, Raj, "High Performance TCP/IP Networking," Prentice Hall India.
- Murthy, Manoj, "Ad Hoc Wireless Networks," Pearson Education.

Additional Books for extra reading:

- CompTIA® Network+® Study Guide, Sixth Edition by Todd Lammle, Jon Buhagiar.
- CompTIA A+ Complete Study Guide: Core 1 Exam 220-1101 and Core 2 Exam 220-1102, 5th Ed. by Quentin Docter, Jon Buhagiar.
- CCNA 200-301 Official Cert Guide, Cisco Press, Wendell Odom.

Additional Note: The students will be required to use Cisco packet tracer to simulate basic network environments and perform basic networking operations. Learning Cisco packet tracer will be a self-guided activity.

S.No.	Description	No. of Lectures
01	Overview of basic concepts in Computer Networks	4-6
	Computer communications: salient features, Datagram, circuit, and connection-oriented networks, Multiple access: contention and ordered techniques, TCP/IP v6 group of protocols: routing and end-to-end reliability, Routing protocols, IPv4 and IPv6, Subnetting, CIDR, VLAN.	
02	Internet Protocols	6-8
	Point-to-point routing: OSPF, BGP, Performance Analysis of multicast and anycast protocols, Reliability Issues in Broadcast and Multicast. TCP and its variants. Performance Issues in TCP: Modelling and Analysis. Active Queue Management in TCP Networks	
03	Wireless Networks	5-7
	Wireless Communication Basics, Wave propagation models, wireless physical layer, Antenna. Link Layer - Wireless Medium Access Control, MAC protocol survey. IEEE 802.11 and variants, Low Power MACs, IEEE 802.15.4, ZigBee. TCP for Wireless Internet.	
04	Traffic Management	5-7
	QoS definition, Mapping models, Integrated and Differentiated Services Architecture. Label Switching, MPLS.	
05	Software Defined Networking	5-7
	Switching architectures, Data, control, and management planes, Dynamic tables, autonomous	

	switching, SDN architecture, Network function virtualization, Open flow, SDN implementations	
06	Some Advanced and Additional Topics	4-6
	Introduction to Data Center Networking, Characteristics and challenges in Data Center Networking, Topologies and Architecture in Data Center Networking, Information Centric Networking, Named Data Networking	

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X	X					X	X		X